WMT Paper 1

Part 1: Theories

1. Which protocol is responsible for ensuring that emails are synchronized across multiple devices rather than just downloaded locally?
A) SMTP
B) POP3
C) IMAP
D) FTP

2. In the Client-Server interaction model, which statement is fundamentally true regarding the initiation of communication?
A) The server initiates the conversation to provide updates.
B) The client always initiates the conversation.
C) Both can initiate the conversation simultaneously.
D) Protocols determine which side initiates based on traffic.

3. According to the hierarchy of the Domain Name System (DNS), which of the following represents a "2nd level zone"?
A) `.` (Root)
B) `.com` C) `google.com`
D) `drive.google.com`

4. What is the primary difference between Physical and Logical network topologies?A) Physical defines the layout of wires, while Logical defines how media is accessed.
B) Logical defines the layout of wires, while Physical defines media access.
C) They are synonymous in modern Ethernet networks.
D) Physical refers to LANs, while Logical refers to WANs.

5. Which network type is specifically characterized by high speed (100 Mbps to 10 Gbps) and private ownership over a limited geographic area like a campus?
A) WAN
B) MAN
C) PAN
D) LAN

6. What was the original primary purpose for the invention of the World Wide Web by Tim Berners-Lee in 1989?

A) To create a global e-commerce platform.

B) To share research and documents across computers.

C) To replace the ARPANET.

D) To develop a decentralized social media network.

7. In the context of IP addressing, how many total addresses are possible in the IPv4 32-bit scheme?

A) 1.2 billion

B) 3.4×10^{38}

C) 4,294,967,296

D) 65,536

8. Which protocol is described as the "foundation of the web" and operates as a stateless protocol?

A) TCP

B) IP

C) HTTP

D) HTTPS

9. When a browser receives an HTTP response, where is the actual content of the webpage (the HTML) typically located?

A) Status Line

B) Headers

C) Body

D) Meta-tags

10. Which network size typically spans a city or a large campus, covering an area up to 50 km?

A) LAN

B) MAN

C) WAN

D) Internet

Part 2: HTML Code Base

11. What does the `<!doctype html>` declaration communicate to the web browser?

A) The document uses XHTML 1.1 standards.

B) The document is an HTML5 document.
C) The document is encrypted.
D) The document contains JavaScript.

12. When creating a description list, which tag is used specifically to provide the "definition" or "value" of a term?

A) `<dl>`
B) `<dt>` C) `<dd>`
D) `<li>`

13. In an HTML table, if you want a single data cell to span across three columns, which attribute should you apply?

A) `rowspan="3"` B) `colspan="3"`
C) `width="3"`
D) `scope="row"`

14. Which HTML5 attribute is used to pair an `<input>` element with a predefined list of options in a `<datalist>`?

A) `id`
B) `name` C) `list`
D) `for`

15. What is the correct way to create an "explicit label" that is linked to a form control for better accessibility?

A) Wrap the `<input>` inside the `<label>` tags.
B) Use the `for` attribute on the `<label>` matching the `id` of the `<input>`.
C) Use the `name` attribute on the `<label>`.
D) Link them using the `class` attribute.

16. Which form method is used when the browser tacks the query string onto the end of the URL?

A) POST
B) SEND
C) GET
D) PUT

17. If you use the attribute `multiple` in a `<select>` tag without specifying a `size`, what is the typical default number of visible items?

A) 1
B) 2

C) 4
D) All items

18. Which tag would you use to create a thematic break or a horizontal ruler across a webpage?
A) `<br>` B) `<hr>`
D) `<div>`
D) `<span>`

19. In HTML5, what happens if a browser does not support a specific new input type like `type="date"`?
A) It crashes the page.
B) It ignores the input field entirely.
C) It falls back to `type="text"`.
D) It displays a broken image icon.

20. To make a form input mandatory before submission, which attribute should be added to the tag?
A) `readonly`
B) `autofocus` C) `required`
D) `value`

Part 3: CSS Code Base

21. Using the CSS Box Model, calculate the total width of a `div` with the following properties: `width: 300px; padding: 15px; border: 2px solid; margin: 10px;`
A) 327px
B) 334px *(Calculation: 300 + 152 + 22 + 0)*
C) 354px
D) 317px

22. Which CSS property allows you to include padding and borders within the specified width of an element to prevent "float drop"?
A) `display: block` B) `box-sizing: border-box`
C) `clear: both`
D) `float: none`

23. What is the correct order for styling link pseudo-classes to ensure they function correctly?

A) visited, link, hover, active
B) link, hover, visited, active
C) link, visited, hover, active
D) active, hover, visited, link

24. Which selector would target an `<em>` tag ONLY when it is nested directly inside a `<p>` tag?

A) `p, em` B) `p em`
C) `p.em`
D) `p > em`

25. In the CSS Box Model, which area is described as being "transparent" and serves as the outermost wrapping of the box?

A) Content
B) Padding
C) Border
D) Margin

26. Which value of the `position` property removes an element from the normal document flow and positions it relative to its first positioned ancestor?

A) static
B) relative
C) absolute
D) fixed

27. To stop elements from wrapping around a floated item and force them to drop below it, which property should be used?

A) `float: none`
B) `display: inline` C) `clear: both`
D) `overflow: hidden`

28. How is the hexadecimal color `#fb0` expanded by a browser into its six-digit form?

A) `#fb0000`
B) `#f0b000` C) `#ffbb00`
D) `#00ffb0`

29. Which property is used to collapse adjacent table cell borders into a single border?

A) `border-style: solid` B) `border-collapse: collapse`

C) `border-spacing: 0`
D) `display: table`

30. Which CSS selector is used to target ALL elements on a webpage simultaneously?
A) `body`
B) `html` C) `*`
D) `all`

Part 4: JavaScript Code Base

31. What is the result of the expression `5 / 10` in JavaScript, and why?
A) 0; because it is an integer division.
B) 0.5; because JavaScript does not have explicit integers and treats numbers as floating-point.
C) 0.5; because the result is automatically cast to a string.
D) Error; because division by a larger number is not allowed.

32. Which of the following describes "hoisting" in JavaScript?
A) Variables declared with `let` are moved to the top of the script.
B) Variables declared with `var` are processed before any code is executed.
C) Function calls are prioritized over variable assignments.
D) The browser "hoists" the script tag to the head of the document.

33. If you declare `const colors = ["red", "blue"];`, which of the following operations will trigger an error?
A) `colors[0] = "green";`
B) `colors.push("yellow");` C) `colors = ["black"];`
D) `colors.pop();`

34. What is the primary difference between `==` and `===` comparison operators?
A) `==` is for numbers; `===` is for strings.
B) `===` checks both value and data type, while `==` only checks value.
C) `==` is faster than `===`.
D) There is no difference in modern JavaScript engines.

35. Which DOM method returns a "Live NodeList" that updates automatically when the document structure changes?
A) `querySelectorAll()`

B) `getElementById()` C) `getElementsByClassName()`
D) `querySelector()`

36. In a `do...while` loop, what is guaranteed to happen regardless of whether the condition is initially true or false?
A) The loop will never execute.
B) The code block will execute at least once.
C) The condition is checked before the first execution.
D) The counter is automatically incremented.

37. When using `addEventListener(event, function, useCapture)`, what does the `useCapture` parameter (if set to `true`) define?
A) The event uses bubbling.
B) The event uses capturing.
C) The event is prevented from propagating.
D) The event listener is only executed once.

38. What happens to existing content on a page if `document.write()` is called after the document has finished loading?
A) The new content is appended to the bottom.
B) It triggers a console warning but does nothing.
C) All prior content is removed and replaced.
D) It only works if the script is in the `<head>`.

39. Which method would you use to change the text inside an element while ensuring that any HTML tags in the string are rendered as plain text rather than HTML?
A) `innerHTML` B) `textContent`
C) `outerHTML`
D) `setAttribute()`

40. If a variable is declared as `var x;` but never assigned a value, what is its data type?
A) Null
B) Empty
C) Undefined
D) String

41. In a `switch` statement, what occurs if a `break` keyword is omitted after a matching case?

A) The code stops immediately.
B) It continues executing the next case regardless of whether it matches (fall-through).
C) It triggers a syntax error.
D) It jumps to the `default` case.

42. Which string method is used to extract a portion of a string by specifying a (start index, end index)?

A) `substr()`
B) `charAt()` C) `substring()`
D) `indexOf()`

43. What is the effect of the `continue` statement inside a loop?

A) It exits the entire loop.
B) It skips the current iteration and continues with the next iteration.
C) It restarts the loop from the beginning.
D) It pauses the loop for a specified time.

44. Which DOM selection method is specifically designed to return only the FIRST element that matches a specified CSS selector?

A) `getElementById()`
B) `getElementsByTagName()`
C) `querySelectorAll()` D) `querySelector()`

45. How do you remove a specific attribute (e.g., `class`) from a retrieved DOM element?

A) `element.deleteAttribute("class")` B) `element.removeAttribute("class")`
C) `element.setAttribute("class", "")`
D) `element.class = null`

46. Which global JavaScript function is used to determine if a value is not a legal number?

A) `Number()`
B) `parseInt()` C) `isNaN()`
D) `isFinite()`

47. What is the "DOM Tree" in the context of web browsers?

A) A list of all files in a directory.
B) A tree-like structure of Objects representing the HTML document.

C) A backup of the server-side database.
D) The sequence of scripts executed during page load.

48. Which property of the `event` object is used in an event listener to access the value of the element that triggered the event (e.g., an input field)?

A) `event.value` B) `event.target.value`
C) `event.source.data`
D) `event.element.id`

49. Why is it considered "better practice" to use `addEventListener()` rather than adding listeners directly in HTML (e.g., `onclick="..."`)?

A) It is faster to execute.
B) It allows for total separation of HTML and JavaScript code.
C) It prevents the use of anonymous functions.
D) It is only supported in modern CSS.

50. What will `console.log(typeof null)` return in JavaScript?

A) "null"
B) "undefined"
C) "object"
D) "string"